PRESEDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT EXAMINATION SYNDICATE
FORM TWO MOCK EXAMINATION, SEPTEMBER-2019.
BASIC MATHEMATICS

TIME: 2:30HRS

INSTRUCTIONS

- This paper consist of section A and B
- Answer all questions in both section
- Mathematical tables may be used
- Calculator, cellular phone are not allowed
- Write your examination number/ name on every page of your answer sheet
- Where necessary use $\boldsymbol{\pi} = \frac{\mathbf{22}}{\mathbf{7}.}$

SECTION A : 60 MARKS).

1. (a) Simplify 4+3(34-6) ÷ $(3x2)$

   (b) If $\frac{2a-b=-3}{6a-5b}$, find the ratio 5a: 2b

   (c) Simplify completely the expression $\sqrt{24}$ $(\sqrt{3}$ $+\sqrt{6}$) - $\sqrt{8}$ $\sqrt{18}$

2. (a) A shopkeeper sold 192 shirts at the price of sh. 5950 each. Estimate how much money he got.

   (b) Add the following

	Dam	m	dm
	12	7	8
	1	4	9
+	3	4	8

   (c) Write the following number 247.1387

   i. Correct to 4 significant figures

   ii. In standard form.

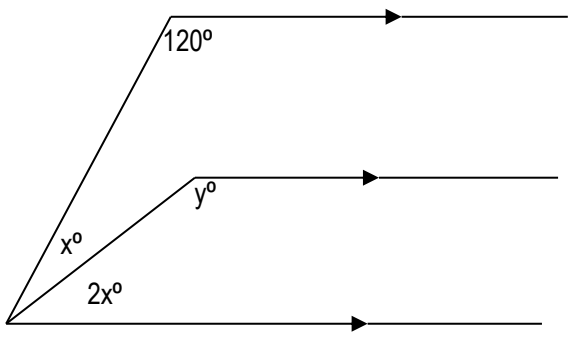
3. (a) Simplify $\frac{125\div0.0025}{8\div0.02}$. Write your answer to 2 decimal places

   (b) A boy spent 35% of his money to buy books. If he was left with 26,000/= find how much money did he have.

   (c) Simplify 4+ $3^1/_2 \div 1^1/_2$ x $^3/_4$

4. (a) Find the number of sides of a regular polygon whose each exterior angle is 15°.

   (b) Determine angle X and Y from figure below.



(c) A piece of wire of length 20cm weight 5g. if a coil of wire of this material weights 1 tonnes. Find the length of wire of coil in km.

5. (a) Find the selling price of an item costing Tshs 24,000/= and sold at a profit of 25%

   (b) Find the solution of the following pairs of linear simultaneous equation graphically.

   2x+y=4

   X + y=3

6. (a) Given that log a=2.4 and log b =1.5 find $\log\sqrt[3]{(b/a)}2$

   (b) Find the value of m

   $(3/4)^m = (4/3)^{2m-1}$

   (c) By using mathematical table calculate 0.000056x5279x0.35

SECTION B (40 MARKS).

7. (a) Find the proportionality ratio need to get a pair of similar triangles if $\overline{CE:DE = 1.2:1,}$ name this pair of simultaneous (similar triangle).

   (b) If p*q=$\frac{p-(p-q)}{p}$ find 2*4

8. (a) The angle of depression of a boat from a light house 25cm high is 18º. How far is the boat from the foot of the light house?

   (b) A translation takes the origin to (3-4). Find where it takes (4,8).

9. (a) In a certain road the number of people got the road accidents in a year where as follows.

Monday	Tuesday	Wednesday	Thursday	Friday
320	310	280	450	250

   Draw a bar chart to illustrate this information.

   (b) Solve the following equations by using the quadratic formula

   i. $X^2 – 4x+ 3=0$

   ii. $X^2 + 3X + 1=0$

10. In a village of 5000 villages, 3300 keeps cows, 1500 keeps goats and 700 don't keep any of those animals. Find the number of villages who keeps both types of animals.